SUMIN PARK

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Google Scholar | LinkedIn | GitHub

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** *Sep 2024 – Present*
  M.S. in Computer Science (expected Feb 2027) Daejeon, South Korea
  - GPA: 4.17 / 4.30
- **Ewha Womans University** *Mar 2016 – Feb 2021*
  B.S. in Brain and Cognitive Sciences & B.S. in Life Sciences Seoul, South Korea
  - GPA: 4.07 / 4.30 (Summa Cum Laude)
  - Thesis: *Differential Mechanism of Positive and Negative Valence Representation: A Focus on Amygdalar Circuitry*

EXPERIENCE

- **CAMAIL, Max Planck Institute for Intelligent Systems / ELLIS Institute Tübingen** *Jul 2026 – Present*
  Research Intern Tübingen, Germany
  - Advisor: T. Konstantin Rusch
  - Research Topic: Looped reasoning for adaptive test-time computation, Blockwise parallelizable linear recurrences for efficient sequence modeling.
- **Big Data Analysis & Learning Laboratory (BigDyL), KAIST** *Sep 2024 – Present*
  Master's Student Researcher Daejeon, South Korea
  - Advisor: Noseong Park
  - Research Topic: Interpretable LLM architectures, Efficient sequence modeling based on linear recurrent structure such as linear attention and state space models (SSMs), mixture-of-experts (MoE) for efficient network specialization
- **Big Data Analysis & Learning Laboratory (BigDyL), Yonsei University** *Jul 2023 – Aug 2024*
  Research Intern Seoul, South Korea
  - Advisor: Noseong Park
  - Research Topic: Graph representation learning, graph neural networks (GNNs)
- **DNI Consulting** *Feb 2022 – Apr 2024*
  Data Scientist Seoul, South Korea
  - Research Topic: Data analysis and modeling for B2B marketing and customer segmentation
- **Cell Physiology Laboratory, Seoul National University** *Aug 2020 – Jan 2021*
  Undergraduate Research Intern Seoul, South Korea
  - Advisor: Seokho Lee
  - Research Topic: Electrophysiology, PFC-involved systems consolidation
  - Investigated how recurrent connectivity between cortical layers (L2/3–L5) in the medial PFC contributes to long-term memory consolidation using patch-clamp electrophysiology.
- **Neurobiology Laboratory, Seoul National University** *Dec 2019 – Feb 2020*
  Undergraduate Research Intern Seoul, South Korea
  - Advisor: Bongkiun Kaang
  - Research Topic: Molecular analysis on dopaminergic reward circuitry

PUBLICATIONS

INTERNATIONAL CONFERENCES

[C7] **Looped Reasoning Finishes Earlier Than You Think** | *Under Review for ICLR 2027*
- **Sumin Park**, and T. Konstantin Rusch

[C6] **Q-Delta: Beyond Key–Value Associative State Evolution** | *ICML 2026*
- **Sumin Park**, Seojin Kim, and Noseong Park

[C5] **STAR: Rethinking MoE Routing as Structure-Aware Subspace Learning** | *ICML 2026*
- **Sumin Park** and Noseong Park

[C4] **How Many Experts Are Enough? Towards Optimal Semantic Specialization for Mixture-of-Experts** | *AAAI 2026*
- **Sumin Park** and Noseong Park

[C3] **Are Graph Transformers Necessary? Efficient Long-Range Message Passing with Fractal Nodes in MPNNs** | *AAAI 2026 (Oral)*
- Jeongwhan Choi, Seungjun Park, **Sumin Park**, Sung-Bae Cho and Noseong Park

[C2] **DARS: Robust Sparse Fine-Tuning with Regularized Subspace Disalignment** | *ICLR 2025 Workshop (SCOPE)*
- **Sumin Park** and Noseong Park

[C1] **PANDA: Expanded Width-Aware Message Passing Beyond Rewiring** | *ICML 2024*
- Jeongwhan Choi, **Sumin Park**, Hyowon Wi, Sung-Bae Cho, and Noseong Park

Invited Talks

- **Quotient Space Convergence for Looped Reasoning** *Nov 17, 2026*
  AI in Scientific Computing: Learning-Based Methods in Scientific Computing, RICAM, Linz, Austria

Industrial Projects

- **Simulation-Based Defect Variable Identification from High-Resolution Images** *Mar 2025 – Dec 2025*
  - **Samsung-KAIST** industry-academia project
  - Developed a multi-task, multi-class image classification framework to predict defective variables from simulation-generated images
  - Designed a ViT architecture with hierarchical attention for efficient high-resolution image processing, achieving 99.41% classification accuracy

Leadership and Extracurriculars

- **Ewha Institute of Brain & Cognitive Science, Ewha Womans University** *Spring 2020 – Fall 2020*
  Member (Spring); President, Biomedical Division (Fall)
- **Debate Association of Ewha (DAE), Ewha Womans University** *Spring 2019 – Fall 2019*
  Regular Member
  - Competed in national debate tournaments (SRT, KNC); SRT 2019 Quarter Finalist

Honors and Awards

- Summa Cum Laude, Ewha Womans University *Feb 2021*
- Dean's List (7 semesters), Ewha Womans University *2016 – 2020*
- Academic Honors Scholarship, Ewha Womans University *Spring 2020*
- Global Leader in Major, Ewha Womans University *Spring 2020*
- Academic Honors Scholarship, Ewha Womans University *Spring 2019*
- Admissions Scholarship (Full Tuition, 2 Years), Ewha Womans University *2016 – 2018*

Skills

- **Programming & Frameworks:** Python, R, C++, PyTorch, Triton
- **Systems & Tools:** Linux, Git, Docker, AWS, SQL
- **Languages:** Korean (native); English (IELTS Academic 8.0, Jun 2026)